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Cytotoxic lanostane triterpenoids from the fruiting bodies of *Piptoporus betulinus*

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ABSTRACT

Chemical investigation of a bioactive methanolic extract of the fruiting bodies of *Piptoporus betulinus* led to the isolation of five previously undescribed lanostane triterpenoids named piptolinic acids A–E, as well as five known lanostane triterpenoids. Their structures were elucidated on the basis of 1D and 2D NMR spectroscopic and HRESIMS analysis. Piptolinic acid A with an unusual moiety (3-hydroxy-4-methoxycarbonyl-3-methylbutyryloxy) at C-3 exhibited comparable cytotoxic activity against human promyelocytic leukemia cell line HL-60 (IC_{50} = 1.77 μM) and human acute monocytic leukemia cell line THP-1 (IC_{50} = 8.21 μM) to those of positive control, fluorouracil (IC_{50} = 6.38 and 4.41 μM, respectively).
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1. Introduction

Piptoporus betulinus (Bull.: Fr.) Karst. (Polyporaceae), commonly known as the birch polypore, is an annual wood-rotting fungus on birch trunks and branches. It is mainly distributed in the Northern regions of Europe, North America, and Asia (Pleszczyńska et al., 2016). *P. betulinus* is an edible mushroom when the fruiting bodies are young, but it was more often used as folk medicine for various medicinal purposes. For example, the fruiting bodies of *P. betulinus* were used to treat various types of cancer in some European countries (Grienke et al., 2014). Pharmacological studies also revealed that the extracts of *P. betulinus* showed various biological activities including anti-proliferation, anti-microbial and anti-inflammation (Cyranka et al., 2011). Previous phytochemical investigation of this fungus has reported some triterpenoids, a hydroquinone and an alkaloid, which exhibited anti-inflammatory, MMP inhibitory, and anti-microbial activities, respectively (Kamo et al., 2003; Wangun et al., 2004; Kawagishi et al., 2002; Schlegel et al., 2000). However, to date, the chemical constituents of the fruiting bodies of *P. betulinus* from China have not been investigated. Moreover, our preliminary experiments revealed that the methanolic extract of *P. betulinus* showed potential cytotoxic activities against several human cancer cell lines, which motivated us to discover cytotoxic compounds from this species. Here we report the isolation and identification of five previously undescribed lanostane triterpenoids named piptolinic acids A–E (**1**–**5**) (Fig. 1), along with five known lanostane triterpenoids (**6**–**10**), and their cytotoxicity evaluation against HL-60 and THP-1 human leukemia cell lines.

2. Results and discussion

Compound **1**, a colorless amorphous solid, showed a molecular formula of $C_{39}H_{60}O_8$ with 10 degrees of unsaturation as determined by HRESIMS at *m/z* 655.4212 $[M-H]^-$ (calcd for $C_{39}H_{59}O_8$, 655.4210). In the ^1H NMR spectrum (Table 1), six tertiary methyl signals (δ_H 0.59, 0.90, 1.02, 1.03, 1.05, and 1.35, each 3H, s), two secondary methyl signals [δ_H 0.96 (d, J = 6.0 Hz), 1.25 (d, J = 7.0 Hz)], two olefinic proton signals (δ_H 5.60, 5.75), a terminal double bond signals (δ_H 4.92, 4.88, each 1H, s), two oxygenated methine protons [δ_H 3.54 (d, J = 5.2 Hz), 4.70 (brs)], as well as two methoxyl signals (δ_H 3.33, 3.63, each 3H, s), were observed. The ^{13}C NMR (Table 1)

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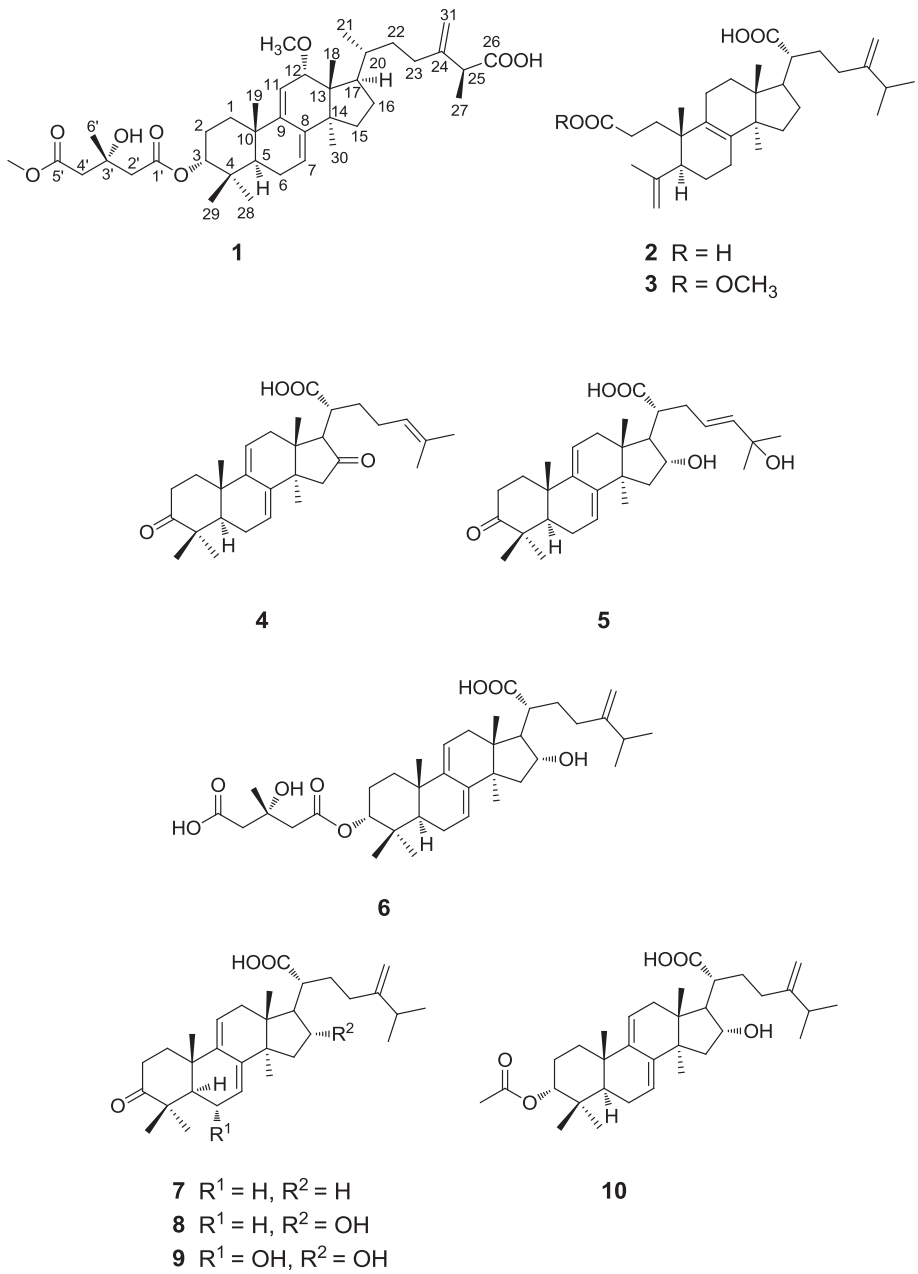

Fig. 1. Structures of compounds **1**–**10**.

spectrum showed 39 carbon signals, which were attributed to 10 methyl, 10 methylene, eight methine (including two sp^2 ones), and 11 quaternary carbons (three of which are characteristic of carbonyl) by HSQC spectra. Aforementioned data suggested that compound **1** is a 24-methyl-lanostane type triterpenoid. Comparison of the NMR data of **1** with those of polyporenic acid A (King et al., 1984) revealed that they had a similar structure. The differences between them are that C-3 of compound **1** is substituted by a functional group other than a hydroxyl group, and the presence of two double bonds except terminal double bond and two methoxyl groups in **1** while only a double bond except terminal double bond and no methoxyl group of polyporenic acid A. A 3-hydroxy-4-methoxycarbonyl-3-methylbutyryloxy moiety was assigned to C-3 on the basis of the HMBC correlations (Fig. 2) from H-3 to C-1′ (δ_C 170.9), from H_3-6′ to C-2′ (δ_C 45.0), C-3′ (δ_C 69.3) and C-4′ (δ_C 44.5), from H-2′ to C-1′, from H-4′ to C-5′ (δ_C 171.6), and from a methoxyl proton signal (δ_H 3.63) to C-5′. Two double bonds ($\Delta^{7,\,9(11)}$ conjugated diene system) were assigned by the ^{1}H-^{1}H COSY correlations of H-6/H-7, H-11/H-12, and the HMBC correlations from H-7 and H_3-19 to C-9 (δ_C 148.6), and from H-11 and H_3-30 to C-8 (δ_C 142.8). A methoxyl group was attached to C-12 by the HMBC correlation from methoxyl signal (δ_H 3.33) to C-12 (δ_C 82.5). The planar structure of compound **1** was thus established. The relative configurations of compound **1** were determined by the NOESY experiment and coupling constant. First, H-3 was determined as a β-orientation based on its broad singlet-like signal (δ_H 4.70, brs). The NOESY correlations from H-12 to H_3-18 and H_3-21, indicated that the 12-MeO took an α-orientation. As the absolute configurations of polyporenic acid A had been assigned unambiguously by the single-crystal X-ray diffraction method, compound **1** shared the same skeleton with polyporenic acid A and its absolute configurations except moiety at C-3 were thus tentatively determined as depicted.

Table 1
^{1}H and ^{13}C NMR data of **1**–**3**.[a]

Position	**1**[b]		**2**[c]		**3**[c]	
	δ_H (mult, *J*, Hz)	δ_C	δ_H (mult, *J*, Hz)	δ_C	δ_H (mult, *J*, Hz)	δ_C
1	1.79 (2H, m)	30.4	1.69 (2H, m)	31.6	1.71 (2H, m)	32.5
2	2.05 (m), 1.77 (m)	22.6	2.48 (m), 1.99 (m)	29.5	2.32 (m), 1.96 (m)	29.4
3	4.70 (brs)	78.3		181.7		174.6
4		36.2		147.6		147.2
5	1.58 (m)	44.2	2.10 (m)	47.4	2.14 (m)	46.8
6	2.07 (2H, m)	22.5	1.79 (m), 1.50 (m)	24.1	1.71 (m), 1.54 (m)	23.9
7	5.60 (brs)	122.6	1.98 (m), 1.37 (m)	26.8	2.00 (m), 1.36 (m)	26.9
8		142.8		139.1		138.6
9		148.6		129.6		129.4
10		37.3		40.4		40.4
11	5.75 (d, 5.2)	113.7	2.08 (m), 1.72 (m)	21.8	2.08 (m), 1.89 (m)	21.6
12	3.54 (d, 5.2)	82.5	1.75 (m), 1.40 (m)	28.6	1.77 (m), 1.42 (m)	29.2
13		47.2		44.2		44.1
14		49.7		50.4		50.4
15	1.68 (m), 1.42 (m)	32.2	1.65 (m), 1.26 (m)	30.7	1.62 (m), 1.25 (m)	30.5
16	2.04 (m), 1.31 (m)	27.0	2.01 (2H, m)	26.0	1.99 (2H, m)	25.9
17	2.28 (m)	43.0	2.10 (m)	47.6	2.13 (m)	47.0
18	0.59 (3H, s)	16.0	0.79 (3H, s)	16.1	0.79 (3H, s)	16.2
19	1.03 (3H, s)	21.5	0.97 (3H, s)	22.6	0.92 (3H, s)	22.2
20	1.41 (m)	36.0	2.30 (m)	47.3	2.30 (m)	47.6
21	0.96 (3H, d, 6.0)	16.5		184.2		181.4
22	1.62 (m), 1.22 (m)	34.3	1.65 (2H, m)	30.8	1.69 (2H, m)	30.9
23	2.19 (m), 2.02 (m)	31.4	1.99 (2H, m)	32.2	2.01 (2H, m)	31.9
24		149.2		155.3		155.1
25	3.12 (q, 7.0)	45.1	2.20 (m)	34.0	2.21 (m)	33.8
26		176.9	0.99 (3H, d, 6.7)	22.0	1.00 (3H, d, 6.5)	21.8
27	1.25 (3H, d, 7.0)	15.4	1.00 (3H, d, 6.6)	21.9	1.01 (3H, d, 6.8)	21.7
28	0.90 (3H, s)	27.1	4.88 (s), 4.67 (s)	113.8	4.89 (s), 4.67 (s)	113.9
29	1.02 (3H, s)	21.4	1.75 (3H, s)	23.0	1.74 (3H, s)	23.0
30	1.05 (3H, s)	27.8	0.96 (3H, s)	25.6	0.93 (3H, s)	25.2
31	4.92 (s), 4.88 (s)	109.7	4.75 (s), 4.67 (s)	106.9	4.75 (s), 4.67 (s)	106.8
12-OMe	3.33 (3H, s)	54.6			3-OMe	
1′		170.9			3.63 (3H, s)	51.5
2′	2.70 (2H, d, 13.4)	45.0				
3′		69.3				
4′	2.70, 2.65 (each 1H, d, 14.9)	44.5				
5′		171.6				
6′	1.35 (3H, s)	26.6				
5′-OMe	3.63 (3H, s)	50.5				

[a] Recorded at 400 or 100 MHz for ^{1}H and ^{13}C.
[b] In CD_3OD.
[c] In $CDCl_3$.

The assignment of 3′*S*-configuration was determined by the comparison of ^{13}C NMR data of C-3′ (δ_C 69.3) with that (δ_C 69.8) of (25*S*,3′*S*)-12α-hydroxy-3α-(3′-hydroxy-4′-methoxycarbonyl-3′-methylbutyryloxy)-24-methyllanosta-8,24(31)-dien-26-oic acid (Kamo et al., 2003), which was isolated from the same species before. Therefore, the structure of compound **1** was elucidated as (3*R*,12*S*,25*S*,3′*S*)-3-(3′-hydroxy-4′-methoxycarbonyl-3′-methylbutyryloxy)-12-methoxy-24-methyllanost-7,9(11),24(31)-trien-26-oic acid, assigned a trivial name piptolinic acid A.

Compound **2** was obtained as a white amorphous powder. Its molecular formula was deduced as $C_{31}H_{48}O_4$ based on the HRESIMS at *m/z* 483.3438 [M-H]$^-$ (calcd for $C_{31}H_{47}O_4$, 483.3474). IR spectrum of compound **2** revealed the presence of carboxyl group with a strong absorption at 1703 cm^{-1}. The ^{1}H NMR spectrum of **2** (Table 1) indicated the presence of four tertiary methyls (δ_H 0.79, 0.96, 0.97, and 1.75, each 3H, s), two secondary methyls [δ_H 0.99 (d, $J = 6.7$ Hz), 1.00 (d, $J = 6.6$ Hz)], and two exomethylene groups (δ_H 4.75, 4.67; 4.88, 4.67; each 1H, s), which were characteristic of 3,4-*seco*-24-methyl-lanostane triterpenoid. Further analysis of ^{1}H, ^{13}C NMR, HSQC, ^{1}H-^{1}H COSY, HMBC and NOESY data allowed us to construct the structure of compound **2**, which possesses a similar structure to that of poricoic acid H (Ukiya et al., 2002). The only difference between them is the absence of a hydroxyl at C-16 in **2**. Thus, the structure of compound **2** was elucidated as depicted, and gave the trivial name piptolinic acid B.

The molecular formula of compound **3** was established as $C_{32}H_{50}O_4$ by the HRESIMS at *m/z* 497.3619 [M-H]$^-$ (calcd for $C_{32}H_{49}O_4$, 497.3631). Comparison of the ^{1}H and ^{13}C NMR data (Table 1) with those of compound **2**, revealed that they shared the same skeleton, the only difference being the presence of an additional methoxyl group (δ_H 3.63, δ_C 51.5), which was assigned to C-3 by the HMBC correlation from methoxyl proton signal to C-3 (δ_C 174.6). The structure of compound **3** was thus determined as depicted and assigned a trivial name piptolinic acid C.

Compound **4**, a white amorphous powder, displayed a molecular formula $C_{30}H_{42}O_4$ based on HRESIMS at *m/z* 465.3001 [M-H]$^-$ (calcd for $C_{30}H_{41}O_4$, 465.3005). The IR spectrum exhibited the characteristic absorption of carbonyl group at 1733 and 1709 cm^{-1}. In the ^{1}H NMR spectrum of **4** (Table 2), seven methyl signals (δ_H 1.05, 1.07, 1.11, 1.12, 1.16, 1.63 and 1.65, each 3H, s), and three olefinic proton signals (δ_H 5.38, 5.41, 5.50), were observed. The ^{13}C NMR (Table 2) and HSQC data revealed the presence of 30 carbon resonances, comprising seven methyl, seven methylene, six methine, and 10 quaternary carbons (three carbonyls). The aforementioned data suggested that compound **4** is a lanostane triterpenoid. Comparison of the ^{1}H and ^{13}C NMR data with those of pinicolic acid B (Rösecke

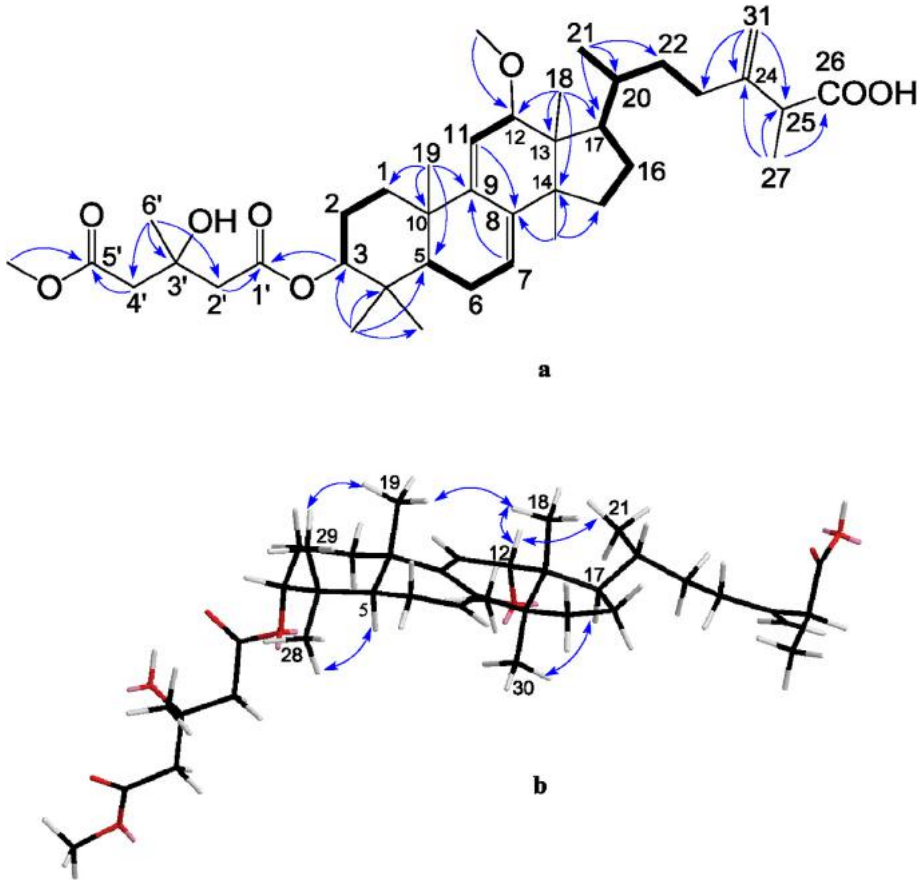

Fig. 2. (a) Key $^{1}H-^{1}H$ COSY (▬) and selected HMBC correlations (H→C) of **1**; (b) Key NOESY correlations (H↔H) of **1**.

and König, 1999), indicated that they were structural analogs and shared the same C-17 side-chain with a Δ^{24}. Further analysis of 1D and 2D NMR data revealed their difference being the position and number of double bond. The $\Delta^{7,\,9(11)}$ conjugated diene system of **4** was assigned by the ^{1}H-^{1}H COSY correlations of H-6/H-7, H-11/H-12, and the HMBC correlations from H-7 and H_3-19 to C-9 (δ_C 143.6), and from H-11 and H_3-30 to C-8 (δ_C 138.7). The structure of compound **4** was therefore elucidated as 3,16-dioxo-lanost-7,9(11),24-trien-21-oic acid, assigned a trivial name piptolinic acid D.

The molecular formula $C_{30}H_{44}O_5$ was determined for compound **5** on the basis of HRESIMS at *m/z* 483.3107 $[M-H]^-$ (calcd for $C_{30}H_{43}O_5$, 483.3110). The ^{1}H NMR spectrum of **5** (Table 2) indicated the presence of seven tertiary methyls (δ_H 0.64, 1.04, 1.07, 1.11, 1.19, 1.22 and 1.23, each 3H, s), four olefinic protons (δ_H 5.42, 5.53, 5.58, 5.64), and an oxygenated methine proton (δ_H 4.06, t, $J = 7.1$ Hz). The ^{13}C NMR (Table 2) and HSQC spectra gave 30 carbon signals comprising seven methyl, six methylene, eight methine (including four sp^2 ones), and nine quaternary carbons (two of which are characteristic of carbonyl). The ^{1}H and ^{13}C NMR data of **5** were similar to those of polyporenic acid C (**8**) (Keller et al., 1996), with the difference between them being the NMR data of the C-17 side-chain. The ^{1}H-^{1}H COSY and HMBC experiment allowed us to determine the structure of C-17 side-chain. The ^{1}H-^{1}H COSY correlations of H-17/H-20, H-20/H-22, H-22/H-23, and H-23/H-24, and the HMBC correlations from H_3-26 (H_3-27) to C-24 (δ_C 139.9), C-25 (δ_C 69.6) and C-27 (C-26), revealed the structure of C-17 side-chain as shown in Fig. 3. The *trans* configuration of Δ^{23} was elucidated by the large coupling constant ($J = 15.2$ Hz) between H-23 and H-24. The relative configuration of H-16 was assigned as β-orientation by the NOESY correlation between H-16 and H-20, while the absolute configurations of compound **5** were determined as depicted based on the biogenetic relationship. Therefore, the structure of compound **5** was elucidated as (16*R*,23*E*)-16,25-dihydroxy-3-oxo- lanost-7,9(11),23-trien-21-oic acid, given the trivial name piptolinic acid E.

Five known lanostane triterpenoids were also isolated and identified as 3-*epi*-(3′-hydroxy-3′-methylglutaryloxyl)-dehydrotumulosic acid (**6**) (She et al., 2012), dehydroeburiconic acid (**7**) (Akihisa et al., 2004), polyporenic acid C (**8**) (Keller et al., 1996), 6α-hydroxypolyporenic acid C (**9**) (Pinhey et al., 1971), and 3-epidehydropachymic acid (**10**) (Tai et al., 1995) on the basis of their spectroscopic data including ^{1}H, ^{13}C NMR and ESIMS.

All of the isolates were tested for their cytotoxicity effects against human promyelocytic leukemia cell line HL-60 and human acute monocytic leukemia cell line THP-1 using the trypan blue method. Compound **1** exhibited comparable cytotoxic activities against HL-60 and THP-1 with IC_{50} values of 1.77 and 8.21 μM, respectively, to those of fluorouracil (IC_{50} = 6.38 and 4.41 μM, respectively). However, the other isolates showed weak or no activity ($IC_{50} > 10$ μM, see Supporting information) against these two cell lines. Preliminary structure–activity relationship (SAR) observation indicated that methoxy groups at C-12 and C-5′ might increase the cytotoxicity activity. However, further SAR studies would be required to confirm this.

3. Conclusion

In summary, phytochemical investigation of a bioactive methanolic extract of the fruiting bodies of *Piptoporus betulinus* led to the isolation and identification of ten lanostane triterpenoids including five previously undescribed ones (**1**–**5**). Compounds **1**–**3**, and

Table 2
^{1}H and ^{13}C NMR data of **4** and **5**.[a]

Position	**4**[b]		**5**[c]	
	δ_H (mult, *J*, Hz)	δ_C	δ_H (mult, *J*, Hz)	δ_C
1	2.11 (m), 1.62 (m)	35.0	2.31 (m), 1.69 (m)	36.3
2	2.76 (m), 2.33 (m)	33.4	2.82 (m), 2.61 (m)	34.3
3		213.5		217.6
4		46.0		47.1
5	1.59 (m)	49.4	1.52 (dd, 12.2, 3.8)	50.9
6	2.11 (m), 2.02 (m)	22.4	2.20 (m), 2.09 (m)	23.2
7	5.50 (d, 6.0)	120.7	5.53 (d, 6.0)	120.3
8		138.7		142.1
9		143.6		144.3
10		36.2		37.0
11	5.41 (d, 5.8)	116.1	5.42 (d, 5.9)	116.8
12	2.73 (m), 2.47 (m)	34.0	2.22 (m), 1.86 (dd, 17.8, 6.4)	35.3
13		41.5		44.1
14		43.0		48.4
15	2.55 (d, 18.5), 2.25 (d, 18.5)	45.3	2.18 (m), 1.45 (d, 13.8)	42.9
16		215.5	4.06 (t, 7.1)	75.9
17	3.03 (m)	56.2	2.11 (m)	55.9
18	1.05 (3H, s)	15.4	0.64 (3H, s)	16.2
19	1.11 (3H, s)	20.5	1.19 (3H, s)	20.9
20	3.00 (m)	43.6	2.42 (ddd, 12.0, 12.0, 3.1)	47.8
21		175.7		177.6
22	3.01 (m), 2.12 (m)	30.5	2.60 (m), 2.26 (m)	34.4
23	2.39 (2H, m)	25.3	5.58 (m)	123.5
24	5.38 (brt, 5.5)	123.4	5.64 (d, 15.2)	139.9
25		130.4		69.6
26	1.65 (3H, s)	24.2	1.22 (3H, s)	28.6
27	1.63 (3H, s)	16.3	1.23 (3H, s)	28.3
28	1.07 (3H, s)	20.9	1.11 (3H, s)	21.4
29	1.16 (3H, s)	24.1	1.04 (3H, s)	24.4
30	1.12 (3H, s)	24.3	1.07 (3H, s)	25.0

[a] Recorded at 400 or 100 MHz for ^{1}H and ^{13}C.
[b] In pyridine-d_5.
[c] In CD_3OD.

6–**10**, are 24-methyl-lanostane triterpenoids, which are well known to be present in the fungal species of the family Polyporaceae, and compounds **2** and **3** possess a 3,4-*seco* structural feature. The cytotoxicities of all of the isolates were tested against human promyelocytic leukemia cell line HL-60 and human acute monocytic leukemia cell line THP-1. Compound **1** showed good cytotoxic activity with IC_{50} values at 1.77 and 8.21 μM, respectively, which suggests that compound **1** may serve as lead structure for anti-leukemia agents.

4. Experimental

4.1. General experimental procedures

Optical rotations were measured on an Autopol VI automatic polarimeter (Rudolph Research Analytical) at room temperature. The IR spectra were recorded on a Nicolet 6700 (Thermo Fisher Scientific) spectrometer. The UV spectra were measured on a Shimadzu UV-2550 UV-visible spectrophotometer. 1D and 2D NMR data were recorded on a Varian 400 MHz instrument with TMS as internal standard. HRESIMS data were acquired using an AB Sciex QTof 5600 + mass spectrometer (AB SCIEX). Semi-preparative HPLC separations were performed on a Hitachi Chromaster system consisting of a 5110 pump, 5210 autosampler, 5310 column oven, 5430 diode array detector and a Phenomenex Luna C_{18} column (250 × 10 mm, S-5 μm), all operated by EZChrom Elite software. All solvents were of ACS or HPLC grade, and were obtained from Tianjin Zhiyuan Chemical (Tianjin, People's Republic of China), Sigma-Aldrich (St. Louis, MO), respectively. Silica gel (300–400 mesh), and C18 reverse-phased silica gel (12 nm, S-50 μm, YMC), were used for column chromatography, and pre-coated silica gel GF254 plates (Qingdao Marine Chemical Plant, Qingdao, People's Republic of China) were used for TLC.

4.2. Fungal materials

The fruiting bodies of *P. betulinus* were collected from the Kanas of Altai Mountains at Xinjiang, China, and identified by Prof. Xinping Yang (Institute of Applied Microbiology, Xinjiang Academy of Agricultural Sciences). A voucher specimen (PB-201501) has been deposited in the Key Laboratory of Plant Resources and Chemistry of Arid Zone, Xinjiang Technical Institute of Physics and Chemistry, Chinese Academy of Sciences (Xinjiang, China).

4.3. Extraction and isolation

The air-dried ground powder of *P. betulinus* fruiting bodies (3.0 kg) was extracted with methanol (25 L × 3) by maceration at room temperature (7 days each time) to afford 755.3 g of crude methanol extract. The crude extract was suspended in distilled water and then extracted successively with petroleum ether (PE), ethyl acetate and *n*-butanol. The ethyl acetate fraction (223.1 g) was subjected to C18 reverse-phased silica gel column chromatography (CC) eluted with MeOH-H_2O (20:80 to 100:0, v/v) in gradient to obtain two fractions A and B. Fraction B (113.6 g) was chromatographed over silica gel CC eluted with PE-EtOAc (100:1 to 0:1, v/v) in gradient to yield eight fractions B1–B8. Fraction B2 (3.0 g) was subjected to silica gel CC eluted with PE-EtOAc (20:1 to 0:1, v/v) in gradient to obtain 13 sub-fractions B2a–B2m. Sub-fraction B2h (53.7 mg) was chromatographed over a column of Sephadex LH-20 and then purified by silica gel CC to yield compound **3** (10.9 mg). Sub-fraction B2k (260.0 mg) was subjected to silica gel CC and then purified by semi-preparative HPLC, eluting with isocratic MeOH-H_2O (85:15, v/v, 3 mL/min) to yield compounds **7** (8.3 mg), **8** (11.7 mg) and **10** (4.1 mg). Fraction B3 (35.4 g) was subjected to silica gel CC eluted with PE-EtOAc (20:1 to 0:1, v/v) in gradient to obtain 11 sub-fractions B3a–B3k. Sub-fraction B3g (3.0 g) was

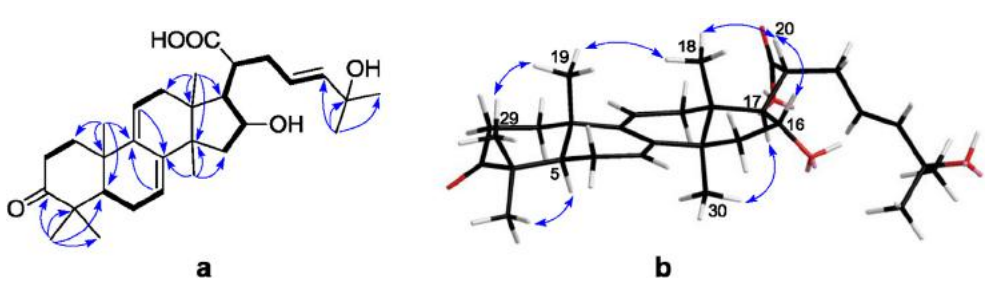

Fig. 3. (a) Key ^{1}H–^{1}H COSY (▬) and selected HMBC correlations (H→C) of **5**; (b) Key NOESY correlations (H↔H) of **5**.

chromatographed over a C18 reverse-phased silica gel CC eluted with MeOH-H_2O (80:20 to 100:0, v/v) in gradient to obtain 12 fractions B3g1–B3g12. Purification of B3g7 (46.2 mg) by semi-preparative HPLC (MeOH/H_2O: 80/20 to 87/13 in 30 min, 3 mL/min) yielded compound **4** (5.3 mg). Fraction B3g11 (652.6 mg) was subjected to silica gel CC and then purified by semi-preparative HPLC (MeOH/H_2O: 95/5 to 98/2 in 24 min, 3 mL/min) to yield compounds **1** (6.2 mg) and **2** (8.2 mg). Fraction B6 (6.2 g) was chromatographed over a C18 reverse-phased silica gel CC eluted with MeOH-H_2O (80:20 to 100:0, v/v) in gradient to obtain 12 sub-fractions B6a–B6l. Sub-fraction B6e (891.3 mg) was purified by silica gel CC to yield compound **9** (10.5 mg) and a fraction B6e1. Purification of B6e1 (22.4 mg) by semi-preparative HPLC, eluting with isocratic MeOH/H_2O (73/27, v/v), yielded compound **5** (6.7 mg). Sub-fraction B6k (255.5 mg) was purified by semi-preparative HPLC eluted with isocratic MeOH/H_2O (83/17, v/v) to yield compound **6** (19.3 mg).

4.3.1. Piptolinic acid A (*1*)

Colorless amorphous solid; $[\alpha]^{20}_D$ 125 (*c* 0.100, MeOH); UV (MeOH) λ_{max} 207, 243 nm; IR ν_{max} 3502, 2962, 1705, 1456, 1375, 1203, 1090, 801 cm^{-1}; ^1H NMR and ^{13}C NMR data, see Table 1; HRESIMS *m/z* 655.4212 $[M-H]^-$ (calcd for $C_{39}H_{59}O_8$, 655.4210).

4.3.2. Piptolinic acid B (*2*)

White amorphous solid; $[\alpha]^{20}_D$ 55 (*c* 0.400, MeOH); UV (MeOH) λ_{max} 220 nm; IR ν_{max} 2958, 1735, 1703, 1455, 1375, 1210, 889 cm^{-1}; ^1H NMR and ^{13}C NMR data, see Table 1; HRESIMS *m/z* 483.3438 $[M-H]^-$ (calcd for $C_{31}H_{47}O_4$, 483.3474).

4.3.3. Piptolinic acid C (*3*)

White amorphous solid; $[\alpha]^{20}_D$ 29 (*c* 0.200, MeOH); UV (MeOH) λ_{max} 218 nm; IR ν_{max} 2959, 1736, 1718, 1455, 1375, 1190, 892 cm^{-1}; ^1H NMR and ^{13}C NMR data, see Table 1; HRESIMS *m/z* 497.3619 $[M-H]^-$ (calcd for $C_{32}H_{49}O_4$, 497.3631).

4.3.4. Piptolinic acid D (*4*)

White amorphous solid; $[\alpha]^{20}_D$ −10 (*c* 0.200, MeOH); UV (MeOH) λ_{max} 208, 242 nm; IR ν_{max} 2920, 1733, 1709, 1452, 1377, 1200, 802 cm^{-1}; ^1H NMR and ^{13}C NMR data, see Table 2; HRESIMS *m/z* 465.3001 $[M-H]^-$ (calcd for $C_{30}H_{41}O_4$, 465.3005).

4.3.5. Piptolinic acid E (*5*)

White amorphous solid; $[\alpha]^{20}_D$ 10 (*c* 0.100, MeOH); UV (MeOH) λ_{max} 204, 242 nm; IR ν_{max} 3367, 2933, 1680, 1450, 1376, 1142, 799 cm^{-1}; ^1H NMR and ^{13}C NMR data, see Table 2; HRESIMS *m/z* 483.3107 $[M-H]^-$ (calcd for $C_{30}H_{43}O_5$, 483.3110).

4.4. Cytotoxicity assay

The cytotoxicity studies of all of the isolates were carried out using the trypan blue method as described previously with suitable modifications. Briefly, cells in logarithmic growth were seeded at 5×10^4 cells/mL in 24-well micro plates and incubated with various concentrations of the compounds and positive control under a humidified atmosphere of 5% CO_2/95% air at 37 °C for three days. The compounds were dissolved in DMSO and then diluted to the proper concentrations. Cell viability was determined after staining the cells with trypan blue. Trypan blue-stained (nonviable) cells and the total cells number were determined with hematocytometer. The growth inhibition in cells after treatment with different concentrations compounds was calculated comparing with control cells (fluorouracil was used here as a positive control), and a half growth inhibitory concentration (IC_{50}) was obtained regression analysis of the concentration response data.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at http://dx.doi.org/10.1016/j.phytochem.2017.07.013.

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